

Asset Pricing Theory: S01

No-Arbitrage, State Prices, and the Stochastic Discount Factor

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Structure of the course

- Email: amedeo.andriollo@dauphine.psl.eu
- Grading follows a 30-70 rule: 70% final exam, 30% project/homework (3rd, 5th sessions TBD).
- Office hours: feel “free” to (DO) come: B612.
- Slides/materials are updated regularly. Report a typo/mistake: corrections help the class. Slides are partially based on last year’s (prof. Imbet).
- This session:
 - No-arbitrage and the Law of One Price; complete markets and state prices; Stochastic Discount Factor; SDF and expected returns.
 - Main reference: Campbell (2017)’s Chp.4.

▷ Next session: consumption-based asset pricing.

What Is (Asset) Pricing?

- An asset is a **resource, owned** by an individual/entity, expected to provide future benefits/costs.
- Assets can take various forms (e.g., physical, such as real estate; intangible, such as patents).
- A **financial asset** is a claim to future, generally uncertain, **cash flows** (e.g., stocks, bonds).
- Abstracting from the institutional name of the security, we represent an asset by:
 - ▷ its price today;
 - ▷ its payoff across possible future states.
- The fundamental question is:

What determines the price today of a random payoff tomorrow?

- There are two complementary ways to attack this question:
 1. **Relative pricing**: which prices are mutually consistent with the absence of arbitrage?
 2. **Equilibrium pricing**: what preferences/risks/technologies/frictions determine the prices?
- Today's lecture begins with the first question and develops the SDF representation.

States of Nature and State-Contingent Payoffs

- Asset pricing is fundamentally about valuing **uncertain future payoffs**.
- A **state of nature** is a complete description of a possible realization of uncertainty at the future date.
- Depending on the model, the state space may be finite, countable, or continuous. We begin with a finite state space for transparency.
- We begin with the simplest benchmark:
 - ▷ two dates, today $t = 0$ and tomorrow $t = 1$;
 - ▷ a finite set of states $S = \{s_1, \dots, s_K\}$;
 - ▷ probabilities $\pi(s) > 0$, with $\sum_{s \in S} \pi(s) = 1$.
- An asset is characterized by its date-1 **state-contingent payoff**:

$$X = (X(s_1), \dots, X(s_K))^T \in \mathbb{R}^K.$$

- Two assets with different names, legal forms, or issuers are economically identical for pricing purposes if they deliver the same payoff in every state.

Reference: (Campbell, 2017, Sec. 4.1.1, p. 83).

Conceptual Route: From Payoffs to Prices

- Having defined assets, payoffs and their associated probabilities, how to study the economy?
↪ Set a utility function, consumption process, production technology, supply-demand equilibrium...
- This session starts before, by asking 3 (sequential) questions:
 1. what can be learned from the implication of the financial markets not offering arbitrage opportunities:

$$\text{Payoffs} + \text{Prices} \xrightarrow{\text{impose NA}} \text{State Prices} \implies \text{SDF}.$$

2. when the available assets span every possible state-contingent payoff:

$$\text{NA} + \text{Completeness} \implies \text{Unique State Prices/SDF}.$$

3. what is the connection between prices to expected returns:

$$1 = \mathbb{E}[MR^g] \implies \text{risk premia depend on covariance with } M.$$

↪ Next session will provide **economic structure** for M using preferences and consumption.

Arbitrage

- There is a widely used expression in finance: “**There is no such thing as a free lunch.**”
- In other words, if something appears to be too good to be true, it probably is.
- **Arbitrage** refers to the practice of taking advantage of price differences in different markets or forms.
- In an efficient market, arbitrage opportunities are quickly exploited and eliminated.
- Early work by Fama during the 1960/70s laid the foundation for modern asset pricing theory and the efficient market hypothesis.

Market Efficiency vs. No Arbitrage

- Zeitgeist about the financial mkts of the 60s/70s: assets react quickly to new info.
 - ▷ Good/bad news (higher/lower future cashflows) tend to **increase/decrease** asset prices.
- The **Efficient Market Hypothesis (EMH)** concerns how information is incorporated into asset prices.
 - ▷ **Weak**: prices incorporate past market information (e.g., prices, volumes).
 - ▷ **Semi-strong**: prices incorporate publicly available information (e.g., news).
 - ▷ **Strong**: prices incorporate both public and private information (e.g., insider info).
- **No Arbitrage (NA)** is a different and substantially weaker restriction:
no trading strategy produces non-negative cash flows at all dates/states and a strictly positive gain somewhere.
- Mind the gap:

No Arbitrage $\neq$ Informational Efficiency.

- We do **not** need the EMH for the results of this session.

Classic reference: Fama (1970)

Portfolios and Payoffs

- Consider N traded assets with today's prices

$$p = (p_1, \dots, p_N)^\top$$

and state-contingent tomorrow's payoff vectors $X^{(1)}, \dots, X^{(N)}$.

- Stack the payoffs in the matrix $A \in \mathbb{R}^{|S| \times N}$:

$$A = \begin{pmatrix} X^{(1)}(s_1) & \dots & X^{(N)}(s_1) \\ \vdots & & \vdots \\ X^{(1)}(s_{|S|}) & \dots & X^{(N)}(s_{|S|}) \end{pmatrix}.$$

- A portfolio is defined by a vector of **holdings** $\theta = (\theta_1, \dots, \theta_N)^\top$.
- Its today's price and tomorrow's payoff vector are:

$$P(\theta) = p^\top \theta, \quad X(\theta) = A\theta.$$

→ Portfolio as **linear** combination of assets.

→ Here θ_i denotes units/holdings, so no adding-up restriction is imposed. If instead we parameterize a fully-invested portfolio by wealth weights w_i , then $\sum_i w_i = 1$.

No Arbitrage

- We define an **arbitrage** as a portfolio θ satisfying:

$$p^\top \theta \leq 0, \quad A\theta \geq 0$$

componentwise, with at least one of these inequalities being strict.

→ Economically, an arbitrage is a trading strategy with **no negative cash flow** at date 0 or in any future state, and with a **strictly positive gain somewhere**.

- There are therefore two economically relevant possibilities:
 1. $p^\top \theta = 0$, while $A\theta \geq 0$ and is strictly positive in at least one state;
 2. $p^\top \theta < 0$, while $A\theta \geq 0$, possibly even $A\theta = 0 \implies$ money today and **zero future obligations**.
- Equivalently:

$$\begin{pmatrix} -p^\top \theta \\ A\theta \end{pmatrix} \geq 0, \quad \begin{pmatrix} -p^\top \theta \\ A\theta \end{pmatrix} \neq 0.$$

Reference: (Campbell, 2017, Sec. 4.2.2, p. 92).

Arbitrage: Three Restrictions

- Spelling out, an arbitrage portfolio θ satisfies three restrictions:

1. **No negative future payoff:**

$$A\theta \geq 0.$$

2. **No positive initial cost:**

$$p^\top \theta \leq 0.$$

3. **A strict gain somewhere:** either the investor receives money today or obtains a strictly positive future payoff in at least one state:

$$-p^\top \theta + \mathbf{1}^\top A\theta > 0.$$

as both terms are non-negative, so their sum is strictly positive $\iff$ at least one gain is strict.

$\hookrightarrow$ This formulation includes both:

$$p^\top \theta = 0, \quad A\theta \geq 0, \quad A\theta \neq 0, \quad \text{and} \quad p^\top \theta < 0, \quad A\theta = 0.$$

Existence of an Arbitrage Opportunity

- An arbitrage exists if there is some portfolio $\theta \in \mathbb{R}^N$ satisfying:

$$A\theta \geq 0, \tag{1}$$

$$p^\top \theta \leq 0, \tag{2}$$

$$-p^\top \theta + \mathbf{1}^\top A\theta \geq \epsilon > 0. \tag{3}$$

- The first two inequalities impose non-negative cash flows.
 - The third inequality converts the strict-gain condition into a weak inequality using an arbitrary $\epsilon > 0$.
 - Because an arbitrage portfolio can be scaled, any arbitrage satisfying a strict inequality can be rescaled to satisfy the chosen normalization ϵ .
- This is a **linear feasibility problem**: finding a feasible θ is equivalent to finding an arbitrage.

Running Example (1/3)

- Consider three future states (Good, Normal, Bad): $S = \{G, N, B\}$.
- In the market, there are two traded assets have payoffs and prices:

	G	N	B	Price
Bond B	1	1	1	0.9
Stock S	2	1	0	0.7

- Consider a claim C with payoff:

$$X_C = (1.5, 1, 0.5)^\top = \frac{1}{2}X_B + \frac{1}{2}X_S.$$

- Hence, if same payoffs must have same prices, its replicating price must be:

$$\frac{1}{2}(0.9) + \frac{1}{2}(0.7) = 0.8.$$

- However, claim C is quoted at $p_C = 0.9$, even though its replicating portfolio costs only 0.8: mkt is mispricing it!

Running Example (2/3)

- Construct the (arbitrage) portfolio: $\theta : +\frac{1}{2}B + \frac{1}{2}S - C$.
- Its cost of today is:

$$p^\top \theta = 0.45 + 0.35 - 0.90 = -0.10.$$

- Its payoff of tomorrow is:

$$A\theta = \frac{1}{2}X_B + \frac{1}{2}X_S - X_C = 0 \quad \text{in every state.}$$

- Therefore the investor receives 0.10 today and has no future liability: $p^\top \theta < 0, A\theta = 0$.
- **Equilibrium and Arbitrage:**
 - Buy 1 bond and 1 stock, and short 2 claims C : eat 0.20 today, zero future liabilities (for free)...
 - ...what stops the investor from scaling this position by 10, 100, or 1000?
- An arbitrage is a deviation from an equilibrium, and excessive demand and supply will quickly adjust prices.

The Law of One Price and the SDF

- Consider two portfolios θ^A and θ^B that generate exactly the same future payoff:

$$A\theta^A = A\theta^B.$$

- The **Law of One Price** states that they must have the same price today: $p^\top \theta^A = p^\top \theta^B$.
- Otherwise, buy the cheaper portfolio and short the more expensive...

No Arbitrage $\implies$ Law of One Price.

- $\hookrightarrow$ The Law of One Price makes the pricing rule **well defined on attainable payoffs**: two portfolios delivering the same payoff vector must have the same price.
- We can therefore write $f(X)$ for the price of an attainable payoff X , independently of the particular portfolio used to replicate it.
 - Given the physical probabilities $\pi(s)$, can this pricing rule itself be represented as an expectation?
- $\triangleright$ To answer this question, we need to introduce risk-neutral probabilities.

Reference: (Campbell, 2017, Secs. 4.1.1 and 4.2.1, pp. 84, 90–91).

Running Example (3/3)

- Suppose the price of C has adjusted to its no-arbitrage value. $\hookrightarrow C$ is redundant, does not expand the payoff span.
- With only B and S :

$$A_{BS} = \begin{pmatrix} 1 & 2 \\ 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

- There are 3 states but:

$$\text{rank}(A_{BS}) = 2 < 3.$$

$\hookrightarrow$ The market is **incomplete**.

- Now introduce an insurance asset I :

$$X_I = (0, 1, 1)^\top, \quad p_I = 0.7.$$

- The new payoff matrix is:

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}, \quad \text{rank}(A) = 3.$$

Complete Markets

- A market is **complete** if any payoff vector $X \in \mathbb{R}^{|S|}$ can be replicated by a portfolio of **traded** assets.
- Formally: the span of traded payoffs $\{X^{(1)}, \dots, X^{(N)}\}$ is the whole space $\mathbb{R}^{|S|}$.
- **Implication:**
For any payoff $X = (X(s_1), \dots, X(s_{|S|}))$, there exist weights $\theta = (\theta_1, \dots, \theta_N)$ such that:

$$X(s) = \sum_{i=1}^N \theta_i X^{(i)}(s), \quad \forall s \in S.$$

Arrow-Debreu Securities

- An **Arrow-Debreu security**, $\delta^{(s)}$, pays 1 in state s and 0 in all other states:

$$\delta^{(s)}(s') = \begin{cases} 1 & \text{if } s' = s, \\ 0 & \text{otherwise.} \end{cases}$$

- If markets are complete, each $\delta^{(s)}$ can be replicated with some portfolio $\theta^{(s)}$.
- Replication of Arrow-Debreu securities under completeness:
 - ▷ Stack payoffs in the matrix $A \in \mathbb{R}^{|S| \times N}$ with entries $[A]_{si} = X^{(i)}(s)$.
 - ▷ **Completeness** $\iff \text{rank}(A) = |S|$ (i.e., full row rank).
 - ▷ For each state s , there exists a portfolio $\theta^{(s)}$ such that

$$A \theta^{(s)} = \delta^{(s)}.$$

- The price of $\delta^{(s)}$ is then called the **state price**:

$$q(s) = \sum_{i=1}^N \theta_i^{(s)} p^{(i)}.$$

From State Prices to Risk-Neutral Probs

- In a complete market, every payoff can be written using Arrow-Debreu securities:

$$X = \sum_{s \in S} X(s) \delta^{(s)}.$$

- By linearity of pricing: $P(X) = \sum_{s \in S} q(s) X(s)$.
- Multiply and divide by the physical probability $\pi(s)$:

$$P(X) = \sum_{s \in S} \pi(s) \frac{q(s)}{\pi(s)} X(s) = \mathbb{E}[MX], \quad M(s) := \frac{q(s)}{\pi(s)}.$$

- The price of a risk-free payoff equal to 1 in every state is: $P_f = \sum_{s \in S} q(s) = \mathbb{E}[M] = \frac{1}{1+r_f}$.
- Define risk-neutral probabilities:

$$\pi^{\mathbb{Q}}(s) = \frac{q(s)}{\sum_{s' \in S} q(s')} = (1+r_f)q(s).$$

- Hence:

$$P(X) = \mathbb{E}[MX] = \frac{1}{1+r_f} \mathbb{E}^{\mathbb{Q}}[X].$$

In-Class Exercise

- Return to our complete three-state economy:

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}, \quad p = \begin{pmatrix} 0.9 \\ 0.7 \\ 0.7 \end{pmatrix}.$$

- Suppose the physical probabilities are: $\pi(G) = \pi(N) = \pi(B) = \frac{1}{3}$.

- Qs:**

1. Verify that the market is complete.
2. Solve $A^\top q = p$ for the state-price vector q .
3. Verify that $q \gg 0$.
4. Compute the SDF $M(s) = q(s)/\pi(s)$.
5. Compute the risk-free rate.
6. Compute the risk-neutral probabilities.

→ See you in 10'. Try to derive each object using the previous slides.

In-Class Exercise: Solution (1/2)

- **1. Is the market complete?**

The payoff matrix is:

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}.$$

Since A is square: $\det(A) = 1 \neq 0$. Hence: $\text{rank}(A) = 3 = |S|$, so mkt is **complete**.

- **2. Solve for state prices.** State prices satisfy: $A^\top q = p$. Therefore:

$$\begin{cases} q_G + q_N + q_B = 0.9, \\ 2q_G + q_N = 0.7, \\ q_N + q_B = 0.7. \end{cases}$$

From the third equation: $q_B = 0.7 - q_N$. Substitute into the first: $q_G = 0.9 - 0.7 = 0.2$. Then $q_N = 0.7 - 0.4 = 0.3$. Lastly, $q_B = 0.4$.

In-Class Exercise: Solution (2/2)

- **3. Check No Arbitrage:** All state prices are strictly positive: $q \gg 0$.
- **4. Compute the SDF.**

Since $\pi(G) = \pi(N) = \pi(B) = 1/3$: $M(s) = \frac{q(s)}{\pi(s)} = 3q(s)$, so: $M = \begin{pmatrix} 0.6 \\ 0.9 \\ 1.2 \end{pmatrix}$.

- **5. Compute the risk-free rate.**

$$P_f = \sum_s q(s) = 0.9 = \frac{1}{1 + r_f}.$$

- **6. Compute risk-neutral probabilities.**

$$\pi^{\mathbb{Q}}(s) = \frac{q(s)}{\sum_{s'} q(s')}, \quad \pi^{\mathbb{Q}} = \begin{pmatrix} 2/9 \\ 1/3 \\ 4/9 \end{pmatrix}, \quad \sum_s \pi^{\mathbb{Q}}(s) = 1.$$

↪ As a final check, the original claim C is priced at:

$$q^{\top} X_C = 0.2(1.5) + 0.3(1) + 0.4(0.5) = 0.8,$$

recovering its no-arbitrage price from Running Example (1/3).

Existence of the SDF

- Under complete markets, we constructed the SDF directly from Arrow-Debreu state prices.
 $\hookrightarrow$ What about incomplete mkts? Does NA alone guarantee the existence of an SDF?
- Consider our one-period economy with payoff matrix $A \in \mathbb{R}^{|S| \times N}$ and today's prices $p \in \mathbb{R}^N$.
- The set of **traded/attainable** payoffs is the linear space:

$$\mathcal{X}_T = \{A\theta : \theta \in \mathbb{R}^N\} \subseteq \mathbb{R}^{|S|}.$$

- The Law of One Price allows us to define a pricing map $f : \mathcal{X}_T \rightarrow \mathbb{R}$ by: $f(A\theta) = p^\top \theta$.
 $\hookrightarrow$ **well defined**: if $A\theta^A = A\theta^B$, by the Law of One Price: $\implies p^\top \theta^A = p^\top \theta^B$.
- Under NA, the pricing map is strictly positive on non-zero non-negative traded payoffs:

$$Y \in \mathcal{X}_T, \quad Y \geq 0, \quad Y \neq 0 \quad \implies \quad f(Y) > 0.$$

- $\hookrightarrow$ **Question**: can this pricing rule be represented by a positive linear functional on the whole payoff space? Can we obtain a state-price vector and so an SDF?

Reference: (Campbell, 2017, Secs. 4.2.1–4.2.2, pp. 90–93).

Step 1: Hyperplane Separation Theorem

- ECONOMIC IDEA:

Arbitrage concerns both the cash flow **today** and payoffs **tomorrow**.

for each traded payoff $Y \in \mathcal{X}_T$, consider the complete net trade: (-cost $\times$ payoff)

$$(-f(Y), Y) \in \mathbb{R} \times \mathbb{R}^{|S|}.$$

- Define the linear subspace of attainable net trades: $\mathcal{Z} = \{(-f(Y), Y) : Y \in \mathcal{X}_T\}$, and the positive cone: $\mathcal{K} = \mathbb{R}_+^{1+|S|}$. Both $\mathcal{Z}$ and $\mathcal{K}$ are convex.
- NO ARBITRAGE: (net) negative costs cannot produce strictly positive payoffs: only the origin.

$$\mathcal{Z} \cap \mathcal{K} = \{0\}.$$

- MATHEMATICAL IDEA:

Since $\mathcal{Z}$ is a linear subspace and $\mathcal{K}$ is the positive cone, under $\mathcal{Z} \cap \mathcal{K} = \{0\}$ the Separating Hyperplane Theorem gives a strictly positive vector $(\lambda, q) \gg 0$ and an associated linear functional $F(a, X) := \lambda a + q^\top X$.

- Because $\mathcal{Z}$ is a **linear subspace**, the separating functional vanishes on $\mathcal{Z}$:

$$F(-f(Y), Y) = -\lambda f(Y) + q^\top Y = 0, \quad \forall Y \in \mathcal{X}_T.$$

Step 2: Pricing Functional and Representation

- Since $\lambda > 0$, normalize the separating vector to $\lambda = 1$, so that:

$$f(Y) = q^\top Y \quad \forall Y \in \mathcal{X}_T, \quad q \gg 0.$$

- Define on the **whole payoff space** $\mathbb{R}^{|S|}$ the linear functional: (for an arbitrary conceivable payoff)

$$\varphi(X) := q^\top X.$$

↪ φ **extends** the observed pricing rule from attainable payoffs to all state-contingent payoffs:

$$\varphi(Y) = f(Y), \quad \forall Y \in \mathcal{X}_T.$$

- Moreover, since $q \gg 0$, the extension is **strictly positive**: $X \geq 0, X \neq 0 \implies \varphi(X) > 0$.
- In finite states, let $\pi(s) > 0$ for every state,

$$\varphi(X) = \sum_{s \in S} q(s)X(s) = \sum_{s \in S} \pi(s) \frac{q(s)}{\pi(s)} X(s).$$

- Define $M(s) := \frac{q(s)}{\pi(s)} > 0$, so that the pricing functional admits the expectation representation:

$$\boxed{\varphi(X) = \mathbb{E}[MX].}$$

Prices as Expectations

- We have moved from No Arbitrage to the SDF in two steps:
 1. **Separation:** NA implies the existence of $q \gg 0$ such that:

$$f(Y) = q^\top Y, \quad \forall Y \in \mathcal{X}_T.$$

2. **Representation:** defining $M(s) = q(s)/\pi(s)$,

$$f(Y) = \sum_{s \in S} q(s)Y(s) = \mathbb{E}[MY], \quad \forall Y \in \mathcal{X}_T.$$

- Hence, for every traded asset $i = 1, \dots, N$: $p_i = \mathbb{E}[MX_i]$.
- Therefore No Arbitrage implies the existence of a **strictly positive SDF** pricing all traded assets.
- Conversely, if $M > 0$ and $p_i = \mathbb{E}[MX_i]$ for every traded asset, an arbitrage cannot exist.
- In our finite-state economy:

$\text{NA} \iff \exists M > 0 \text{ pricing all traded assets.}$

Complete vs. Incomplete Mkts: Uniqueness

- Under NA, the set of positive state-price vectors is: $\mathcal{Q} = \{q \gg 0 : A^\top q = p\}$, and:

$$\text{NA} \iff \mathcal{Q} \neq \emptyset.$$

↪ **No Arbitrage gives existence.**

- If markets are complete: $\text{rank}(A) = |S| \implies \ker(A^\top) = \{0\}$. If $q, q' \in \mathcal{Q}$: $A^\top(q - q') = 0 \implies q = q'$.
↪ **completeness gives uniqueness:**

$\text{NA} + \text{Complete Markets} \implies \text{a unique positive } q \text{ and a unique positive } M.$

- If markets are incomplete, $\ker(A^\top) \neq \{0\}$. If q_0 is one solution, then:

$$q = q_0 + h, \quad h \in \ker(A^\top),$$

also prices all traded assets whenever $q \gg 0$.

Finite vs. Continuous States

- Discrete vs. continuous state space is a **different question** from completeness.
- In our finite-state model:

$$\mathcal{X} = \mathbb{R}^{|S|}, \quad \varphi(X) = q^\top X.$$

- With a continuous state space, payoffs are random variables of a function space: $L^2(\Omega, \mathcal{F}, \mathbb{P})$.
- If $\varphi : L^2 \rightarrow \mathbb{R}$ is a continuous linear pricing functional, the **Riesz Representation Theorem** implies that there exists a unique $M \in L^2$ such that:

$$\varphi(X) = \langle M, X \rangle = \mathbb{E}[MX] \quad \forall X \in L^2.$$

- Positivity of the pricing functional translates into $M \geq 0$ a.s.; strict positivity implies $M > 0$ a.s.
- **Completeness** still means that every admissible payoff in the chosen payoff space can be generated by traded strategies. Infinite dimensionality makes the functional analysis richer: does not imply incompleteness.

Projection Geometry of the SDF (1/2)

- Consider the finite-state case and define the probability-induced inner product:

$$\langle X, Y \rangle = \sum_{s \in S} \pi(s) X(s) Y(s) = \mathbb{E}[XY].$$

- Let the **span of traded payoffs** be: $\mathcal{X}_T = \{A\theta : \theta \in \mathbb{R}^N\}$.
- Any SDF M admits the orthogonal decomposition:

$$M = M^* + M_{\perp}, \quad M^* \in \mathcal{X}_T, \quad M_{\perp} \in \mathcal{X}_T^{\perp}.$$

- M^* is the projection of M onto the traded payoff space:

$$M^* = \text{proj}_{\mathcal{X}_T}(M).$$

$\hookrightarrow M_{\perp} \in \mathcal{X}_T^{\perp}$ is the orthogonal part: $\langle M_{\perp}, Y \rangle = 0$ for any $Y \in \mathcal{X}_T$.

$\hookrightarrow$ For every **traded payoff** $Y \in \mathcal{X}_T$: $\mathbb{E}[MY] = \mathbb{E}[M^*Y]$.

Projection Geometry of the SDF (2/2)

- For every $Y \in \mathcal{X}_T$:

$$\langle M, Y \rangle = \langle M^* + M_\perp, Y \rangle = \langle M^*, Y \rangle + \underbrace{\langle M_\perp, Y \rangle}_{=0} = \langle M^*, Y \rangle.$$

- All SDFs sharing the same projection M^* generate the exact same prices for traded assets:

$$M_1 = M^* + M_\perp^{(1)}, \quad M_2 = M^* + M_\perp^{(2)} \quad \implies \quad \mathbb{E}[M_1 Y] = \mathbb{E}[M_2 Y] \quad \forall Y \in \mathcal{X}_T.$$

- IMPORTANT DETAIL:

- Uniqueness? In incomplete markets, M^* is the **unique** payoff-space **representer** of the pricing functional, $\mathbb{E}[M^* Y] = P(Y), \forall Y \in \mathcal{X}_T$.
- Sign? M^* need not itself be positive. NA guarantees that at least one strictly positive SDF exists, but that positive SDF need not belong to $\mathcal{X}_T$.

$\hookrightarrow$ Asset prices pin down $M^* = \text{proj}_{\mathcal{X}_T}(M)$, so the part of M in $\mathcal{X}_T^\perp$ is invisible to the pricing equations.

Reference: (Campbell, 2017, Secs. 4.2.1–4.2.2, pp. 91–93).

From the SDF to Expected Returns

- Consider a traded asset i , with today's $p_i > 0$, tomorrow's payoff X_i , **gross** return: $R_i^g := X_i/p_i$.
- Dividing the fundamental pricing equation $p_i = \mathbb{E}[MX_i]$ by p_i :

$$1 = \mathbb{E}[MR_i^g].$$

- For the risk-free asset, define $R_f^g := 1 + r_f$. Assuming R_f^g is constant:

$$1 = R_f^g \mathbb{E}[M] \quad \implies \quad \mathbb{E}[M] = \frac{1}{R_f^g}.$$

- For a risky asset:

$$1 = \mathbb{E}[M]\mathbb{E}[R_i^g] + \text{Cov}(M, R_i^g).$$

- Hence:

$$\mathbb{E}[R_i^g] - R_f^g = -R_f^g \text{Cov}(M, R_i^g).$$

↪ LHS: “premium” over the risk-free, RHS: covariance with SDF.

What Is Risk in Asset Pricing?

- An asset is not risky merely because its payoff is volatile.
↪ What matters for its risk premium is **when it pays**.
 - If an asset pays particularly well when M is high:
 - ▷ it delivers wealth in states in which wealth is especially valuable;
 - ▷ investors value it highly today;
 - ▷ its equilibrium expected return is therefore relatively low.
 - Conversely, an asset that performs badly when M is high provides little insurance against valuable/bad states:
 - ▷ investors require a higher expected return (premium) to hold it.
- ↪ To understand risk premia economically, we must understand **what makes M high or low**.

Where Does the SDF Come From?

- SDF has been obtained from prices and NA. What about microfoundations?
- Consider an investor choosing consumption:

$$\max_{\{C_0, C_1(s)\}} u(C_0) + \beta \sum_{s \in S} \pi(s) u(C_1(s))$$

subject to:

$$C_0 + \sum_{s \in S} q(s) C_1(s) = W_0.$$

- The first-order condition for consumption in state s is:

$$q(s) u'(C_0) = \beta \pi(s) u'(C_1(s)).$$

- Can we find M in terms of consumption levels?

▷ **Next session: Consumption-Based Asset Pricing.**

What Have We Learned?

- Starting only from portfolios and state-contingent payoffs:

$$\text{No Arbitrage} \iff \exists q \gg 0 : p = A^\top q.$$

- State prices give the SDF:

$$M(s) = \frac{q(s)}{\pi(s)}, \quad p_i = \mathbb{E}[M X_i].$$

- If a risk-free asset exists:

$$\mathbb{E}[M] = \frac{1}{R_f^g}, \quad p_i = \frac{1}{R_f^g} \mathbb{E}^\mathbb{Q}[X_i].$$

- Completeness implies uniqueness: $\text{rank}(A) = |S| \implies q$ and M unique.
- Risk premia satisfy:

$$\mathbb{E}[R_i^g] - R_f^g = -R_f^g \text{Cov}(M, R_i^g).$$

- Utility maximization suggests: $M_{t+1} = \beta u'(C_{t+1})/u'(C_t)$.

▷ **Next: Which consumption risks generate observed asset prices?**

Thanks for your attention! See you next time!

Bibliography I

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